

오일러는 도형을 보고 무슨 생각을 할까?

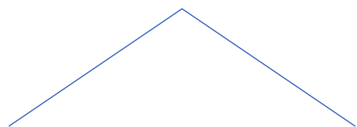
- 그런데 위상을 섞은

수학과 김민국

서론

수학은 분류!

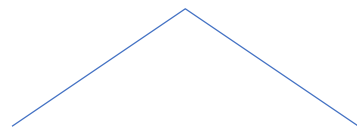
사람



남자

여자

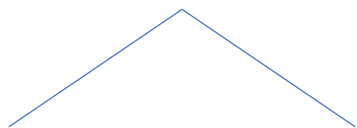
자연수



홀수

짝수

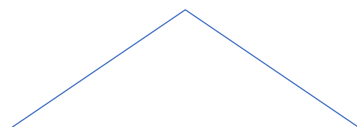
집합



유한
집합

무한
집합

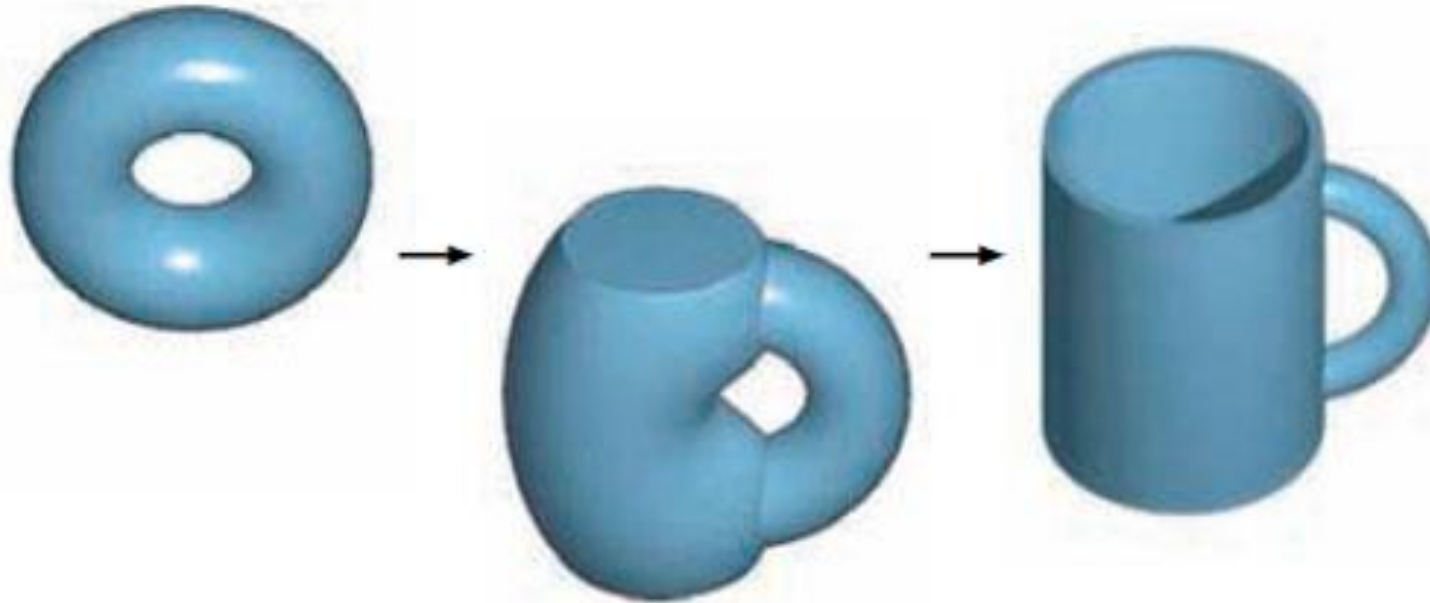
함수



연속
함수

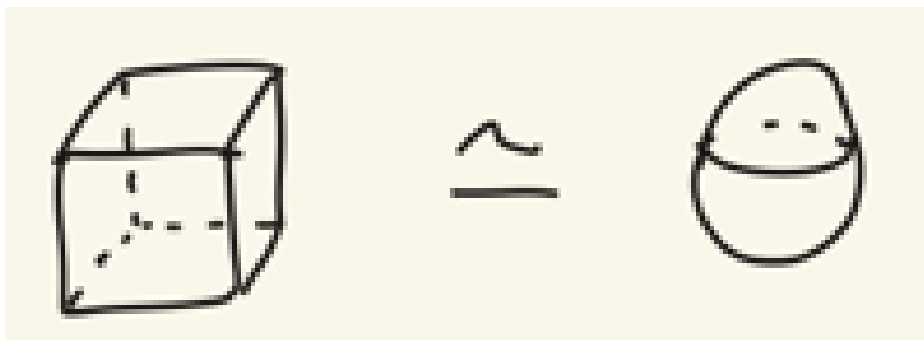
불연속
함수

도넛이 컵으로 변하는 과정



도넛과 머그컵은 위상동형이다!

위상동형의 예시들



도형
(공간)

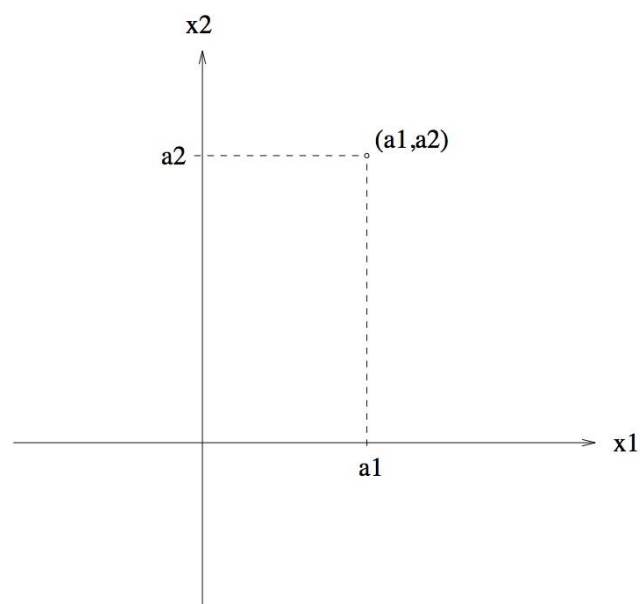
구멍이
있다

구멍이
없다

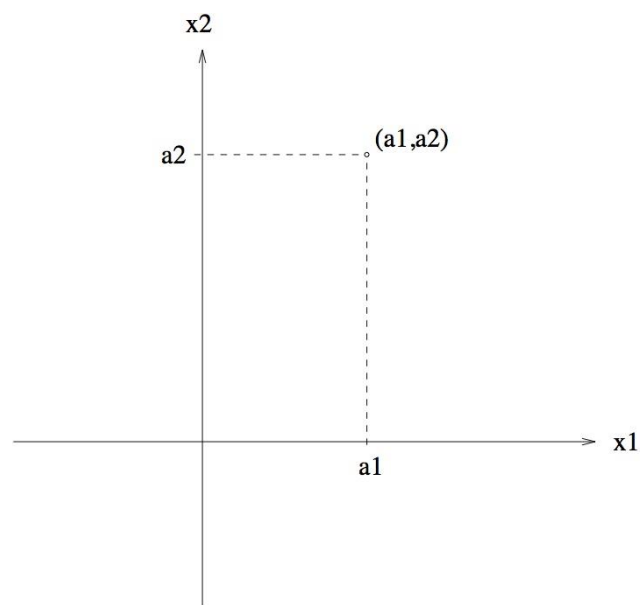
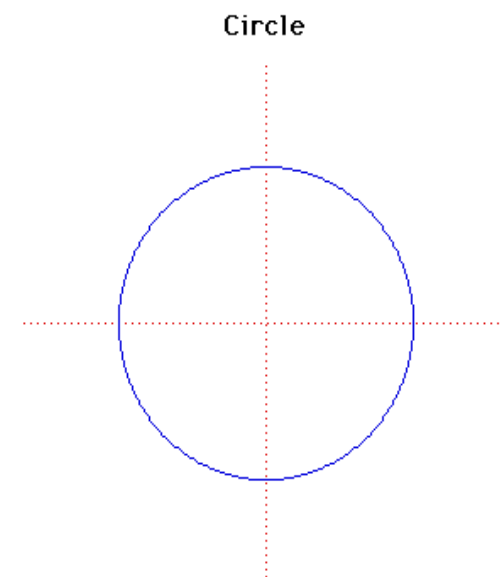
구멍
1개

구멍
2개

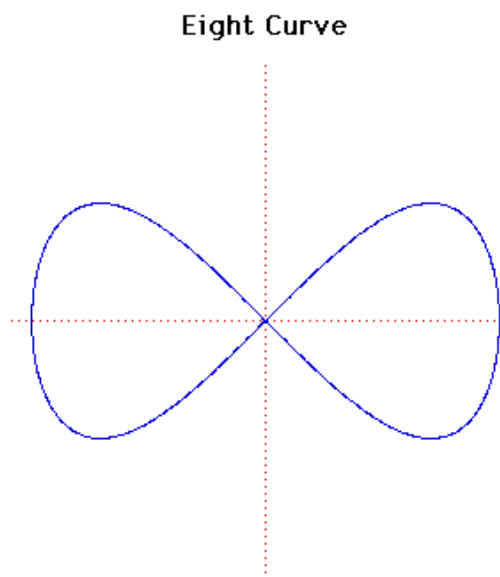
...

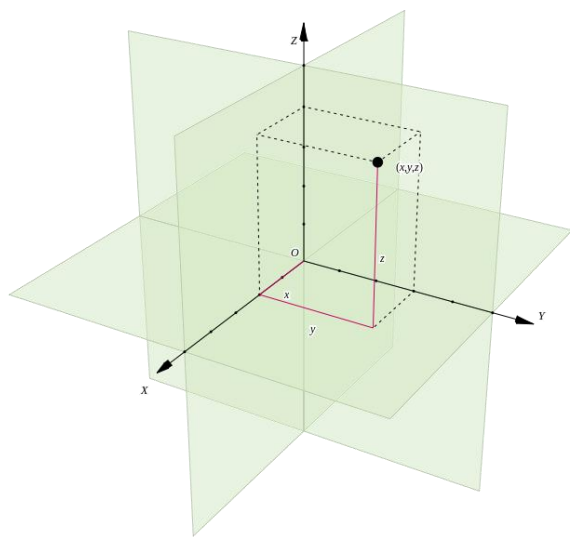


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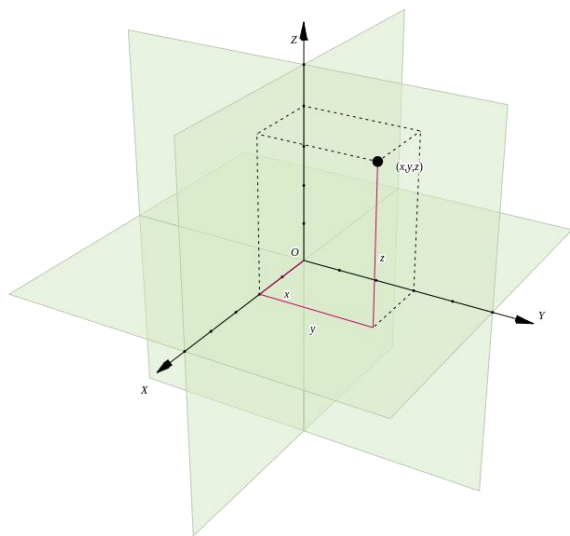
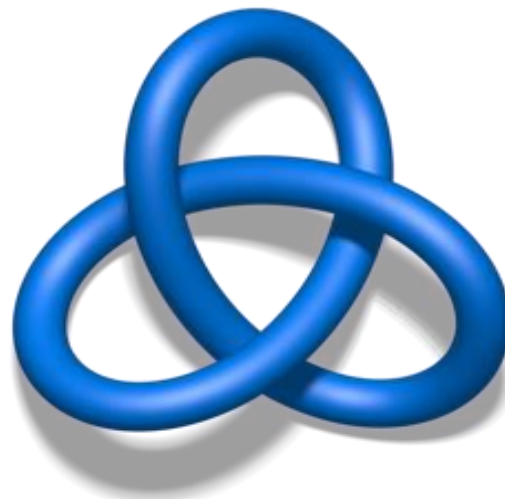


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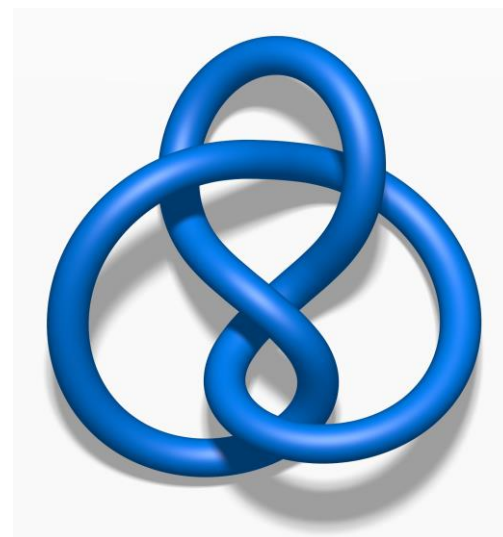




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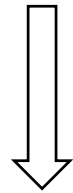


둘은 위상동형이 아니다!

그런데 어떻게 알아내지?

위상적 불변량

A와 B가 위상동형이면, 어떤 값이 같다.



대우

어떤 값이 다르면, A와 B는 위상동형이 아니다.

예시

- 차원
- 구멍의 개수
- 덩어리의 개수
- 경계의 유무
- 특정한 숫자
- 등...

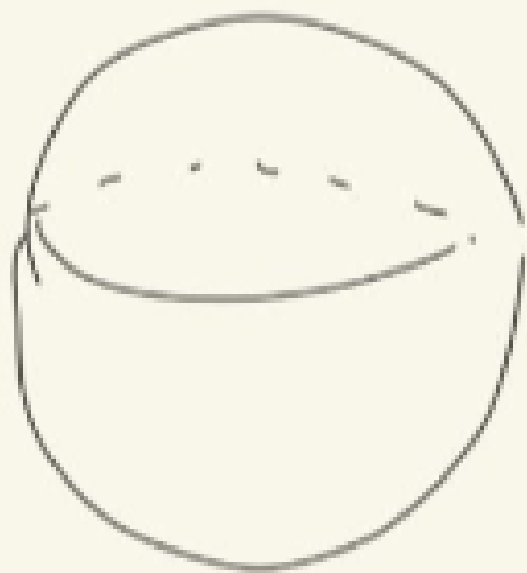
오늘의 위상적 불변량

Euler characteristic

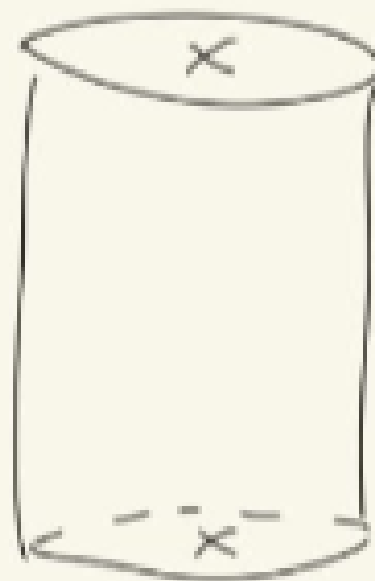
(특정한 숫자)

참고할 점

1. 하나의 덩어리만 고려
2. Orientable만 고려
(몹시우스의 띠 같은 도형들을 고려 X)
3. 경계가 없는 도형만 고려
(앞으로 계속 나아갈 수 있어야함)



O



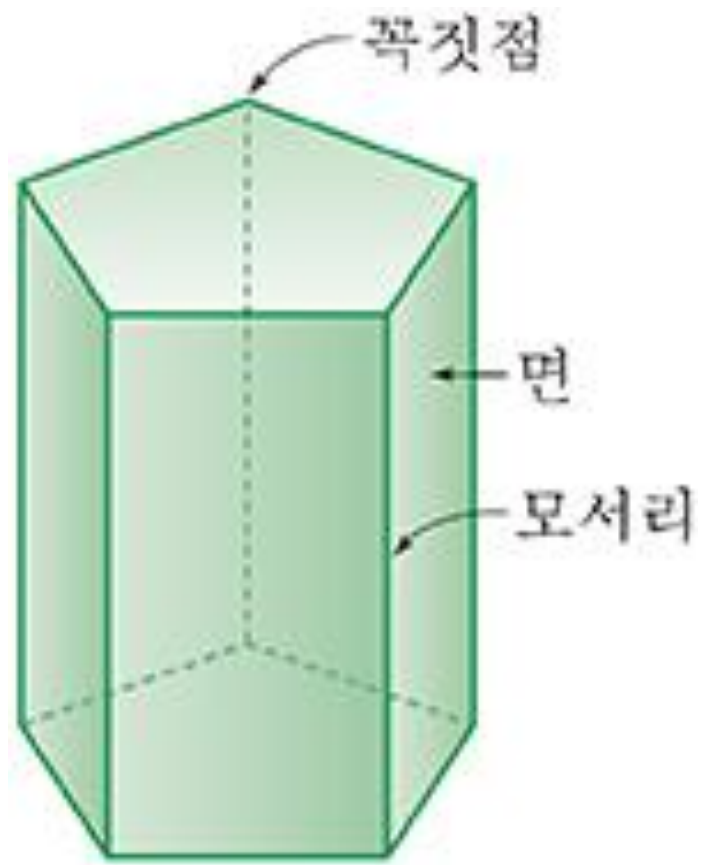
X

목차

- 1) 꼭짓점 - 모서리 + 면 = 2 (오일러의 다면체 공식)
- 2) 꼭짓점 - 모서리 + 면 $\neq$ 2
- 3) Euler characteristic

1. 꼭짓점 - 모서리 + 면 = 2
(오일러의 다면체 공식)

다면체



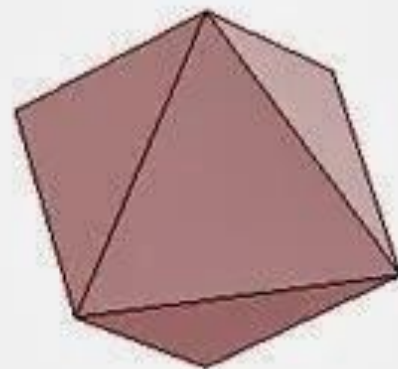
다면체



Dodecahedron



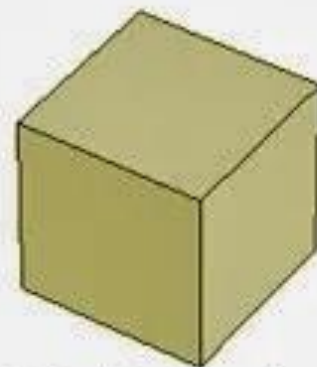
Tetrahedron



Octahedron



Icosahedron



Cube (hexahedron)



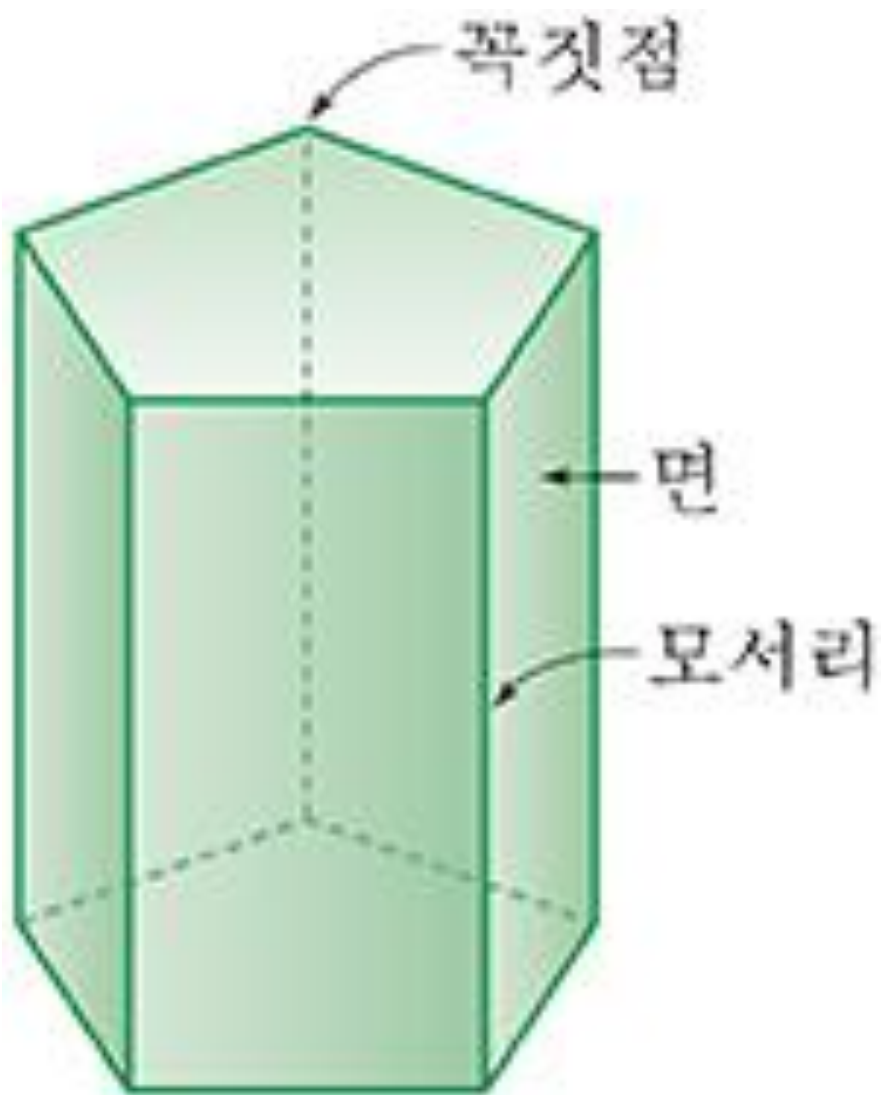
Euler(1707 – 1783)

오일러의 볼록 다면체 공식

$$\text{꼭짓점} - \text{모서리} + \text{면} = 2$$

$$V - E + F = 2$$

(Vertex / Edge / Face)



$$V = 10$$

$$E = 15$$

$$F = 7$$

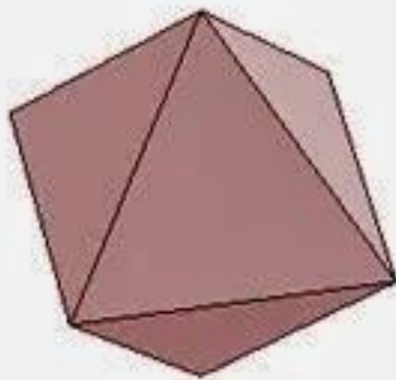
$$V - E + F = 2$$



Dodecahedron



Tetrahedron



Octahedron



Icosahedron



Cube (hexahedron)

$$\text{정4면체} : 4 - 6 + 4 = 2$$

$$\text{정6면체} : 8 - 12 + 6 = 2$$

$$\text{정8면체} : 6 - 12 + 8 = 2$$

$$\text{정12면체} : 20 - 30 + 12 = 2$$

$$\text{정20면체} : 12 - 30 + 20 = 2$$

과연 모든 볼록 다면체에서 성립할까?

STEP 1

가장 처음에 있을 n-각형

$$V = n$$

$$E = n$$

$$F = 1$$

$$\Delta V - E + F = 1$$

N각형

ex)

$$V = 4$$

$$E = 4$$

$$F = 1$$

$$\Delta V - E + F = 1$$



STEP 2

m-각형을 추가할 때

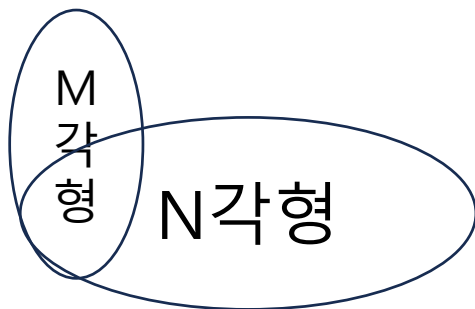
(k개의 모서리를 공유)

$$V = m - (k + 1)$$

$$E = m - k$$

$$F = 1$$

$$\Delta V - E + F = 0$$



ex)

$$V = 4 - (1 + 1)$$

$$E = 4 - 1$$

$$F = 1$$

$$\Delta V - E + F = 0$$



STEP 3

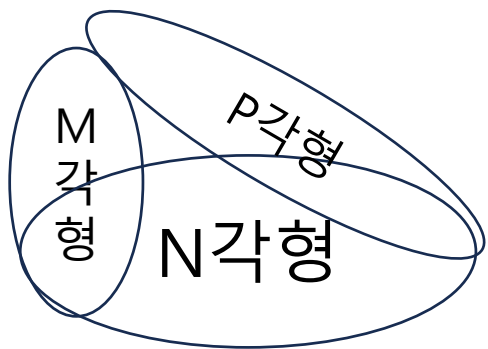
마지막 p-각형으로 덮음

$$V = 0$$

$$E = 0$$

$$F = 1$$

$$\Delta V - E + F = 1$$



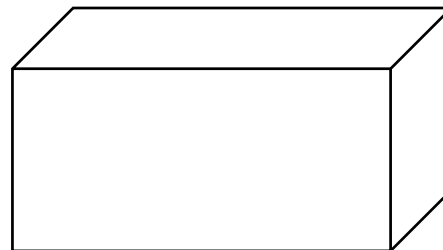
ex)

$$V = 0$$

$$E = 0$$

$$F = 1$$

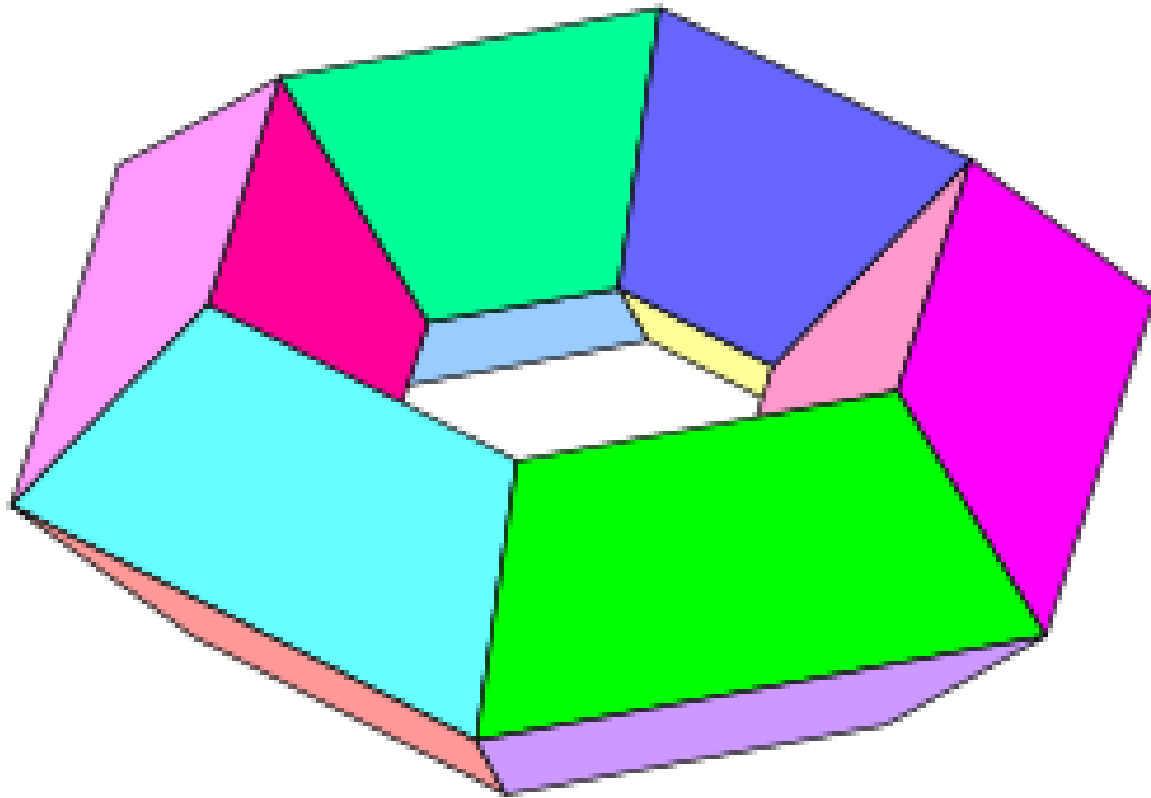
$$\Delta V - E + F = 1$$



최종적으로 $V - E + F = 2$

2. 꼭짓점 - 모서리 + 면 $\neq$ 2

그러나...

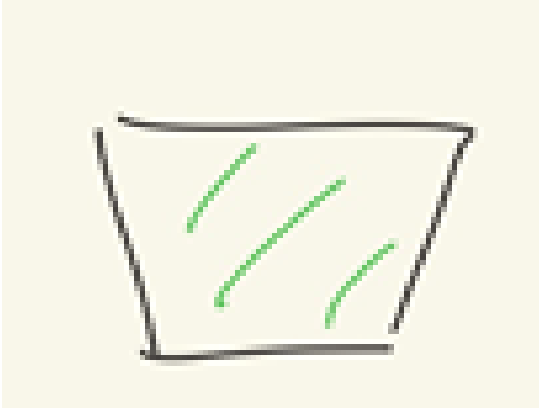


$$V = 24$$

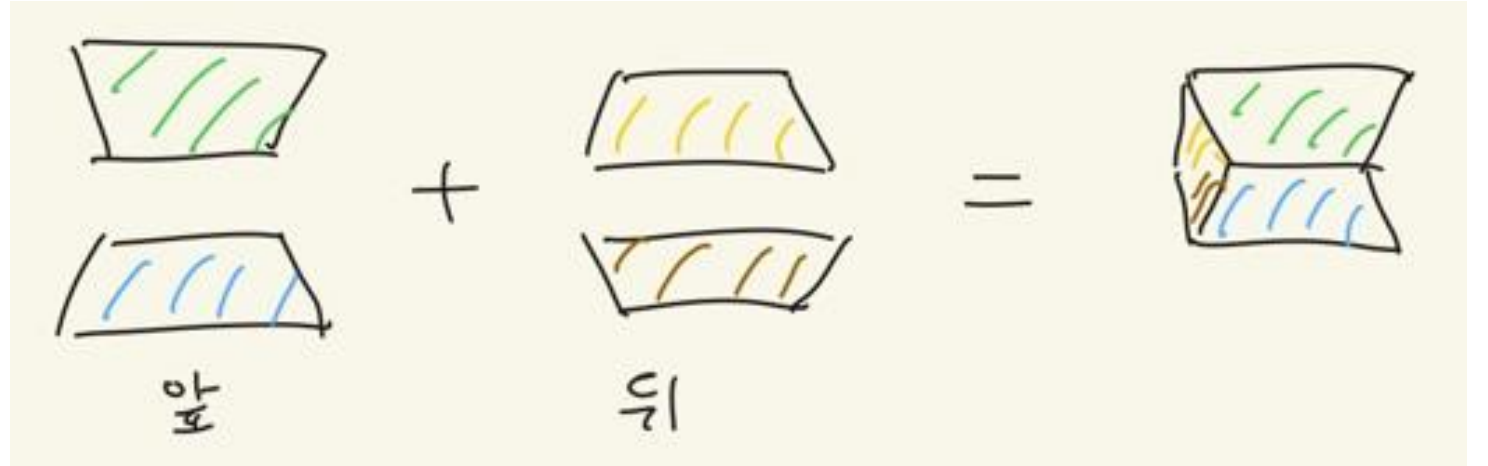
$$E = 48$$

$$F = 24$$

$$V - E + F = 0 ??$$



$$1) V - E + F = 1$$

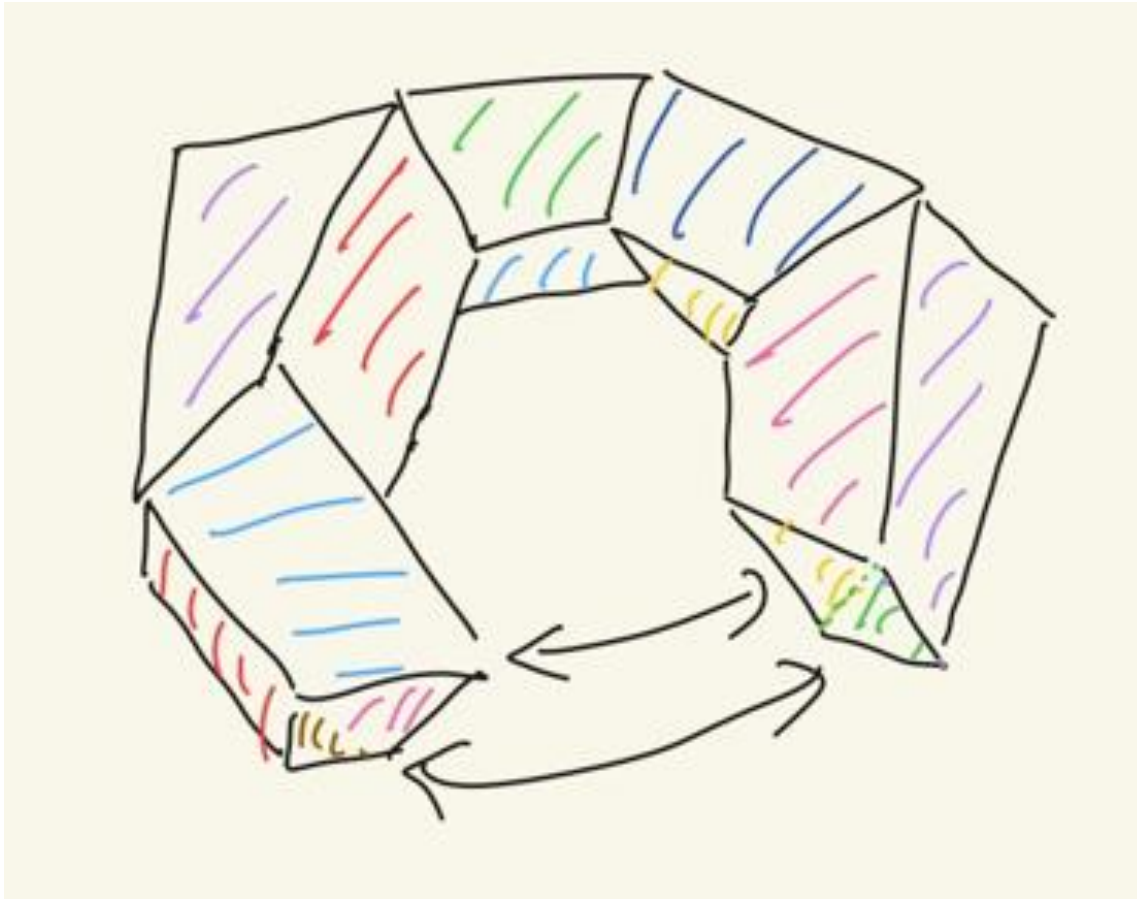


$$2) \Delta V - E + F = 0$$

$$3) \Delta V - E + F = 0$$

$$4) \Delta V - E + F = -1$$

$$\text{총 } V - E + F = 0$$



붙이기 전

$$V - E + F = 0$$

붙인 후

$$V = -4$$

$$E = -4$$

$$F = 0$$

$$\Delta V - E + F = 0$$

붙이면서 닫을 때마다

$\Delta V - E + F$ 은 -2

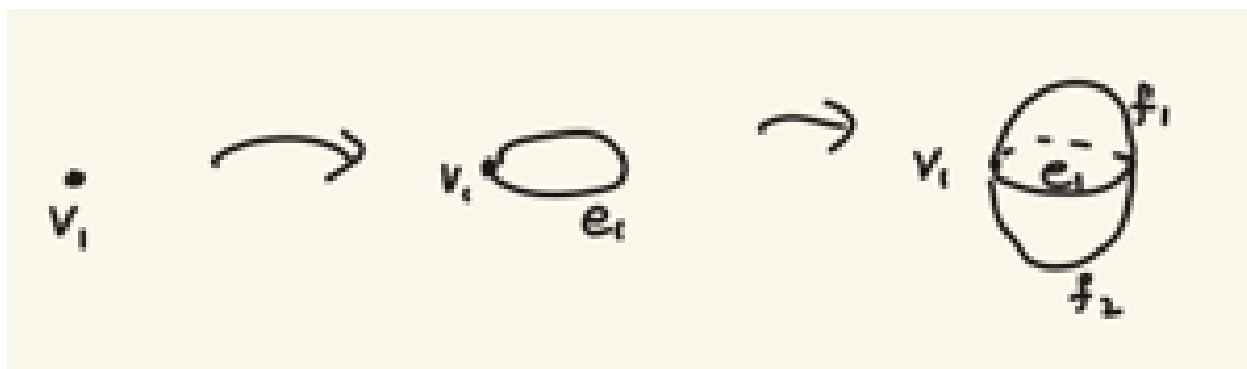
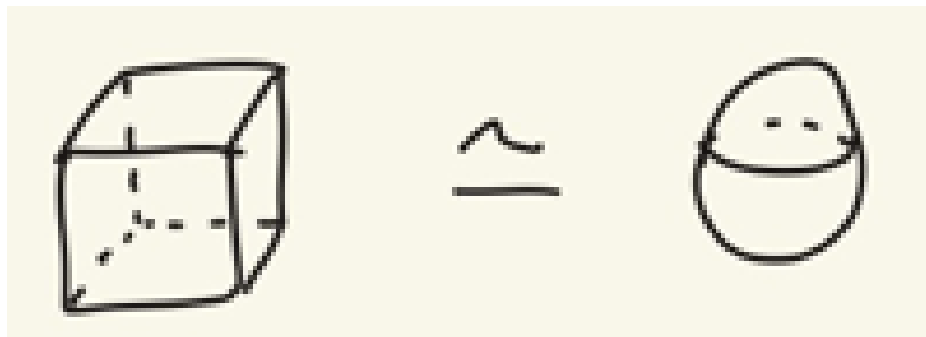


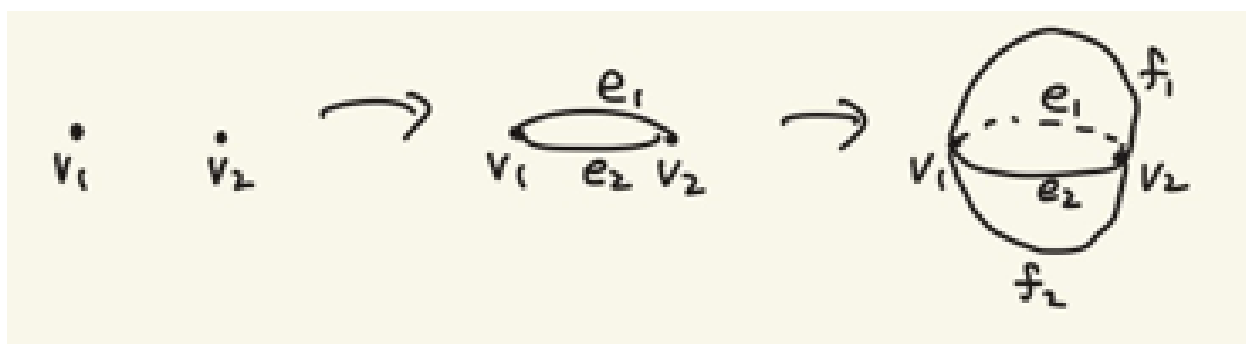
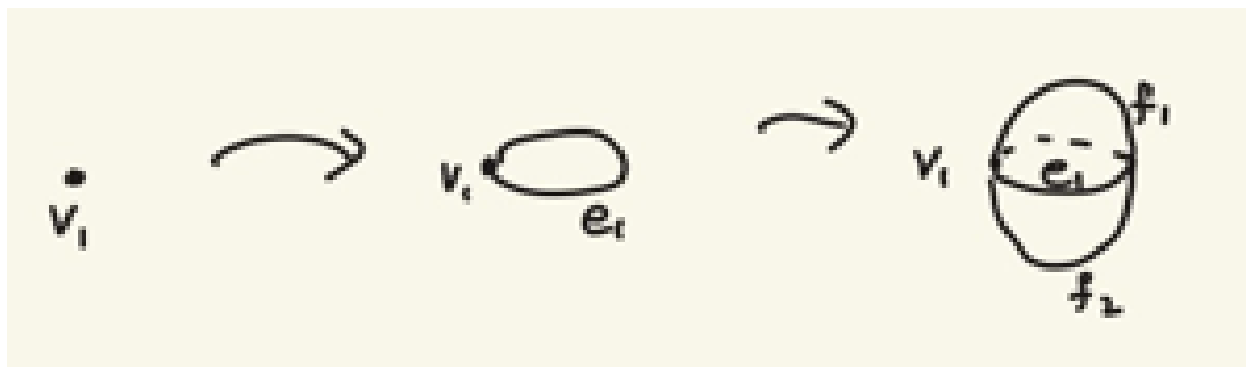
구멍 1개 당

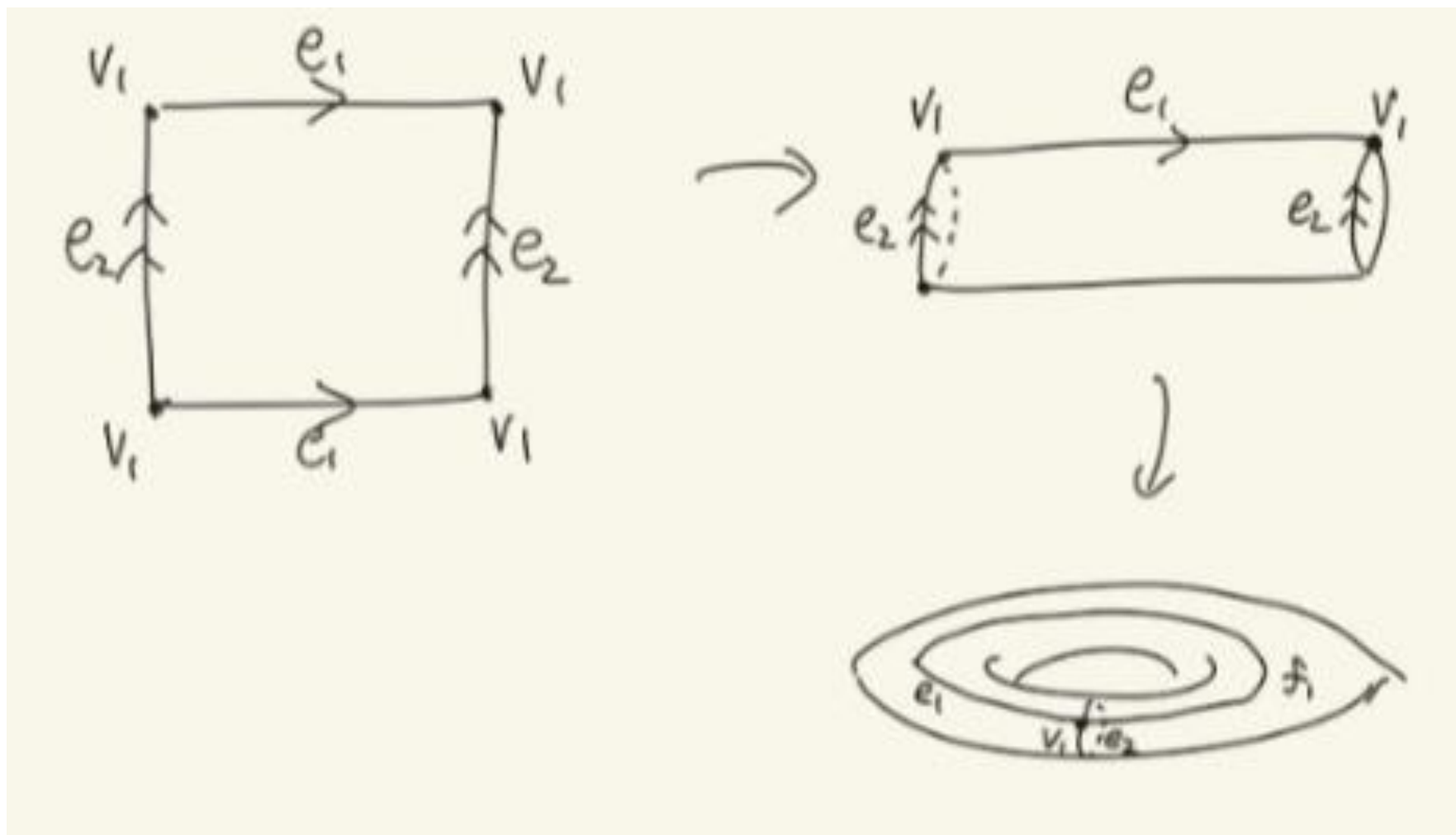
$\Delta V - E + F$ 은 -2

$$V - E + F = 2 - 2g$$

다면체를 벗어나자

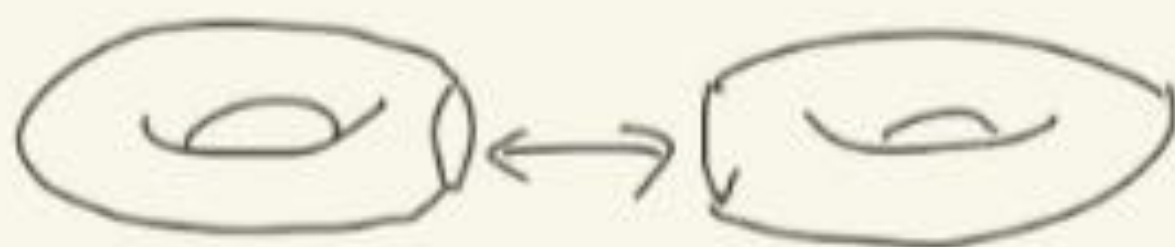






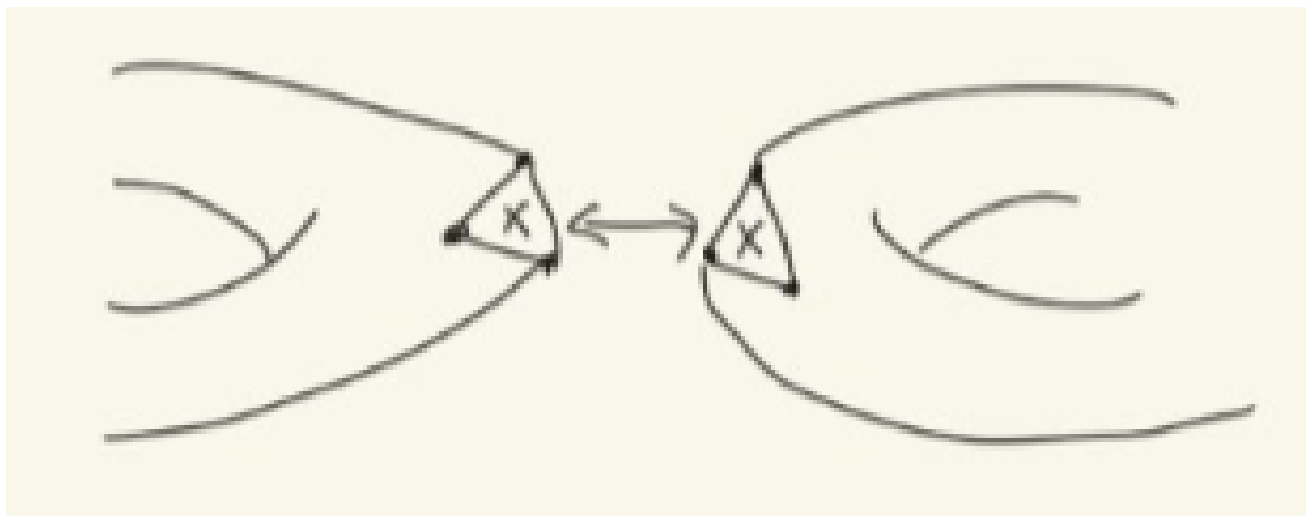
$$V = 1, E = 2, F = 1$$

$$V - E + F = 0$$



$$V - E + F = 0$$

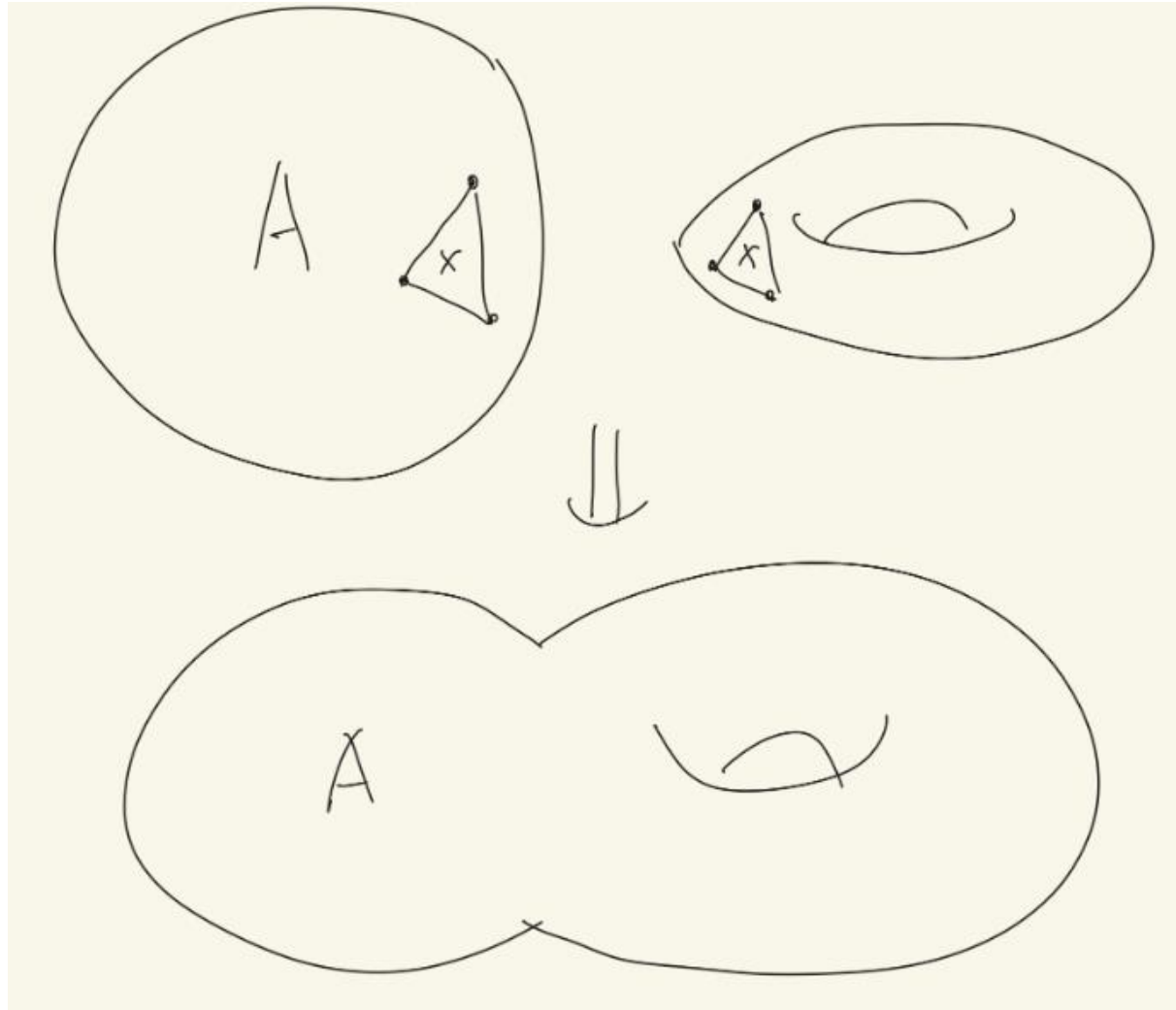
$$V - E + F = 0$$



$$\Delta V - E + F = (-3) - (-3) + (-2) = -2$$

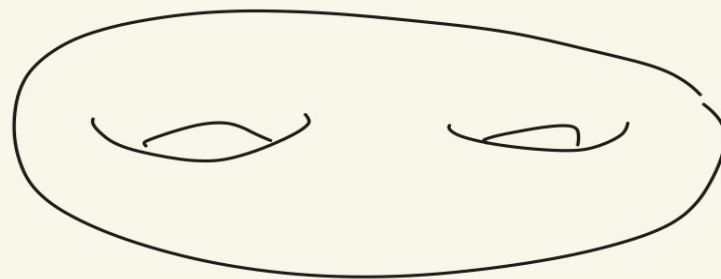
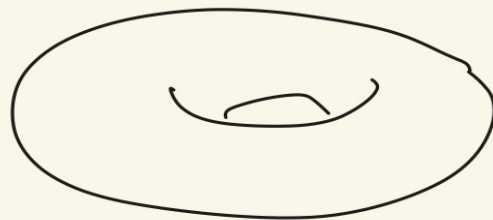
$$V - E + F = n$$

$$V - E + F = 0$$



$$V - E + F = n - 2$$

$$V - E + F = 2 - 2g$$

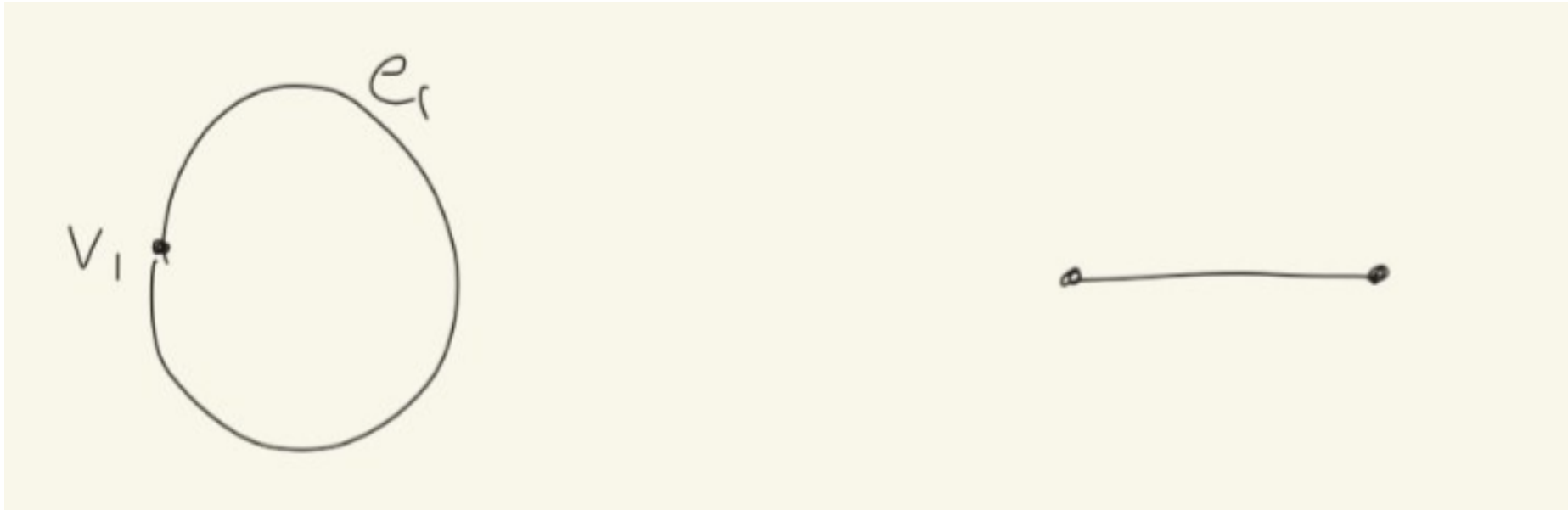


...

g	0	1	2	...
v_{eff}	2	0	-2	...

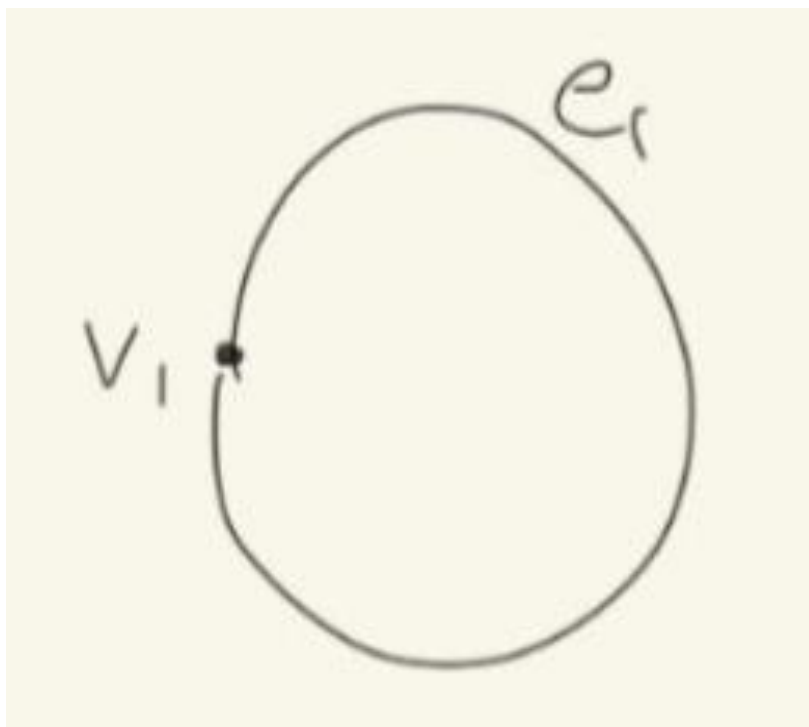
3. Euler characteristic

2차원을 벗어나자!

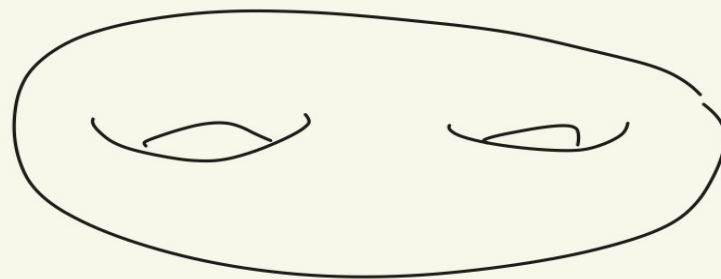
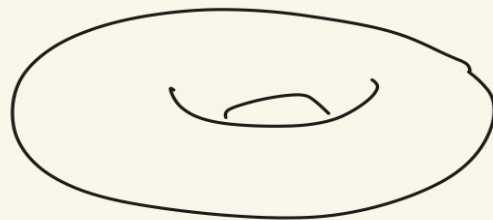


O

X



$$V - E = 0$$



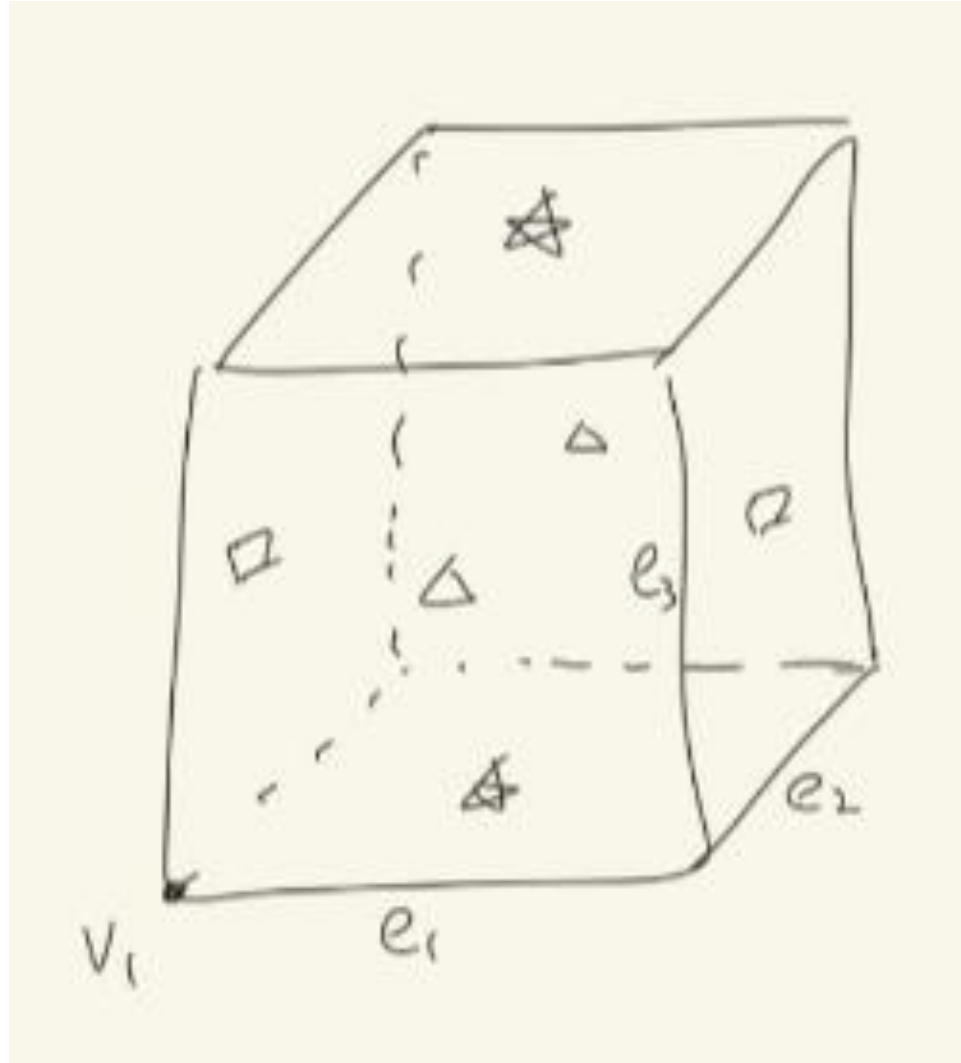
...

g	0	1	2	...
v_{eff}	2	0	-2	...

3차원



X



$$V - E + F - C = 1 - 3 + 3 - 1 = 0$$

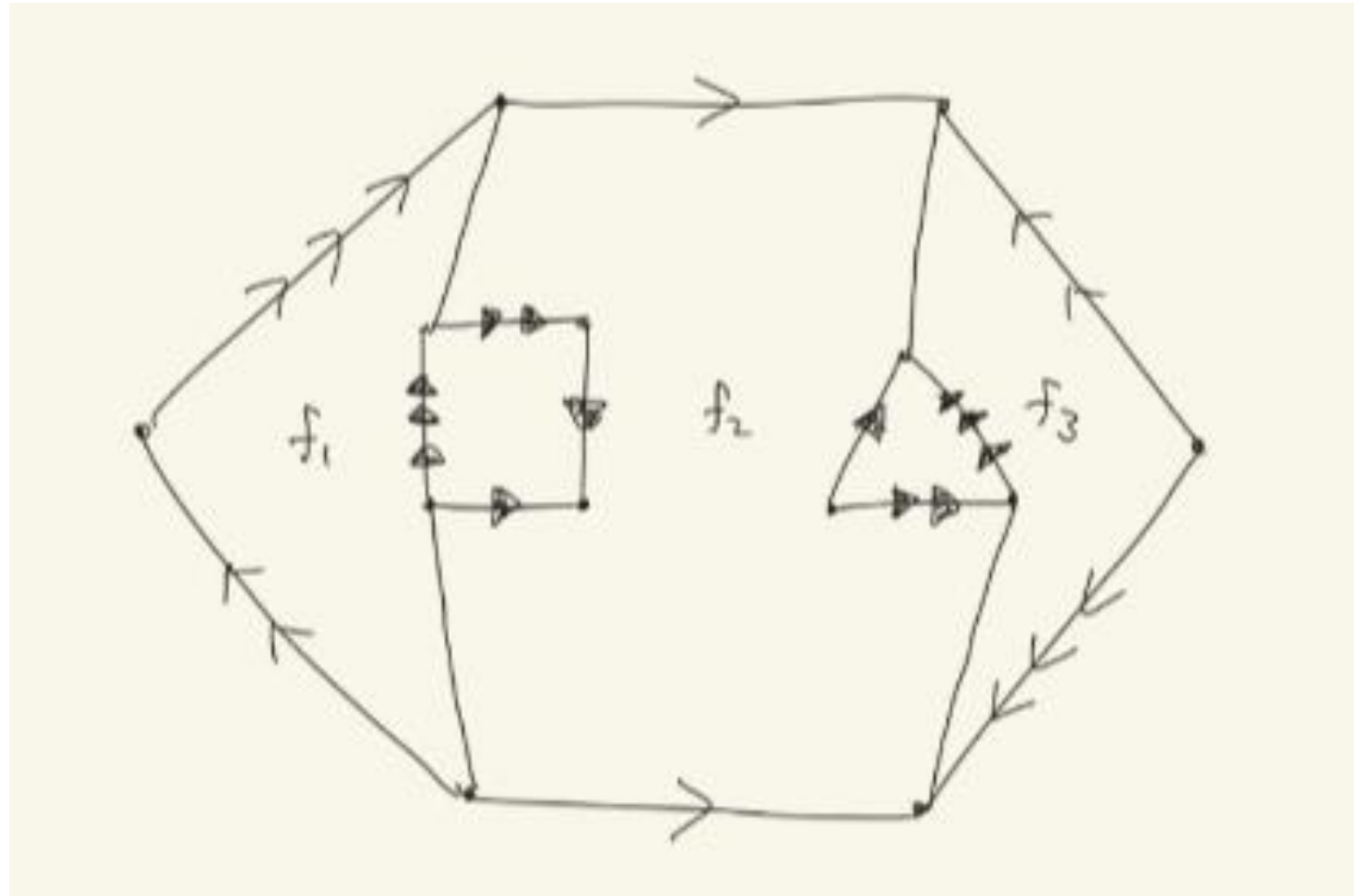
Euler characteristic

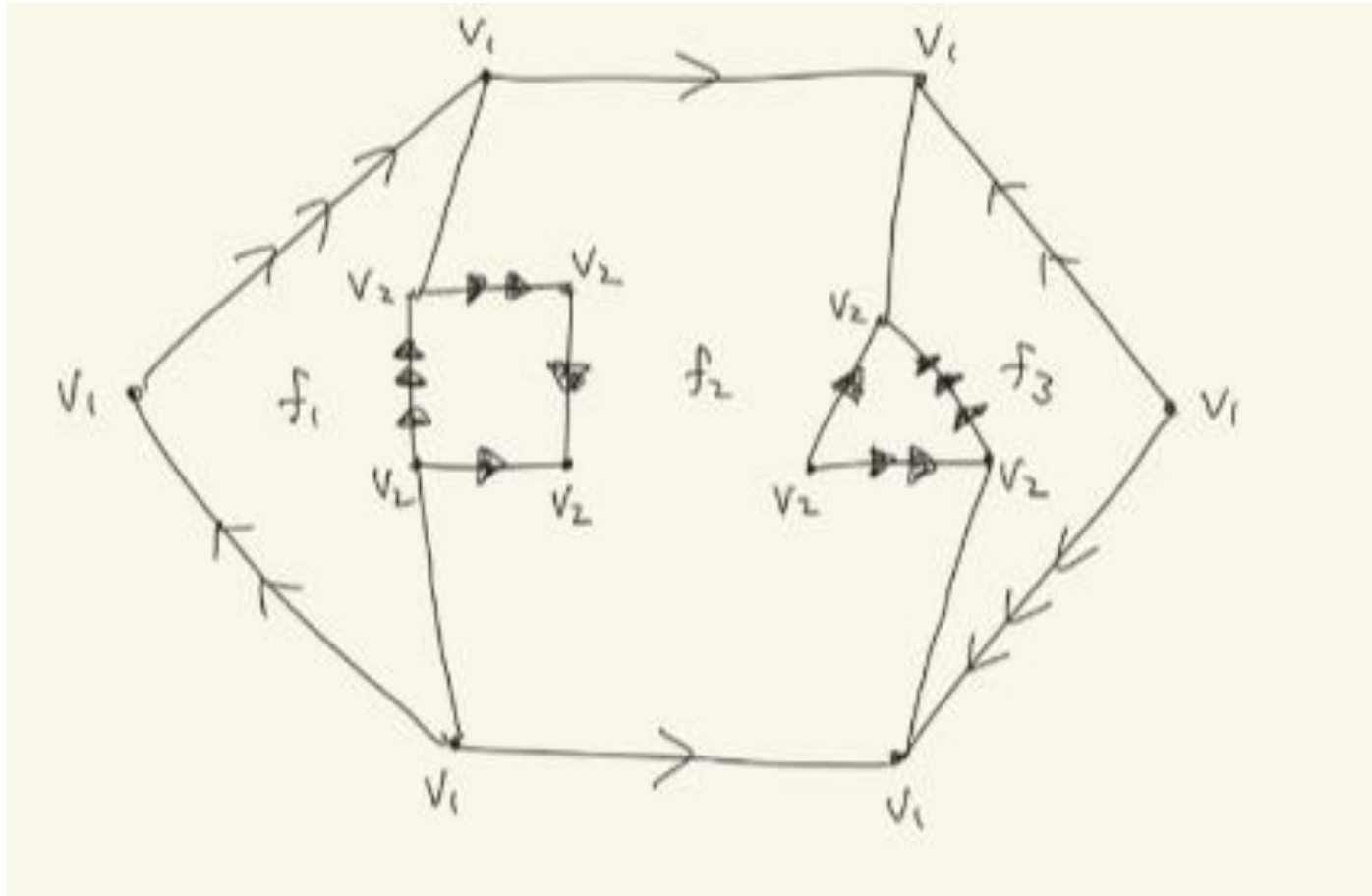
$$n_1 - n_2 + n_3 - n_4 + \dots$$

Euler characteristic은 위상적 불변량!

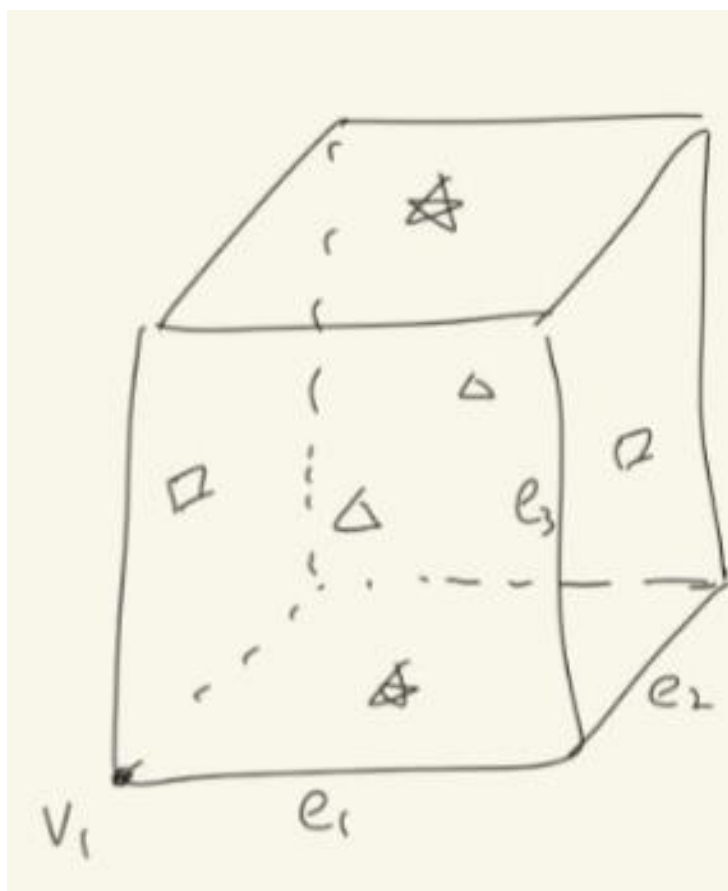
1. 위상동형이면 Euler characteristic은 같다
(공간을 다른 점, 선, 면 조합으로 만들어도 Euler characteristic은 같음)
2. Euler characteristic이 다르면 위상동형이 아니다

퀴즈

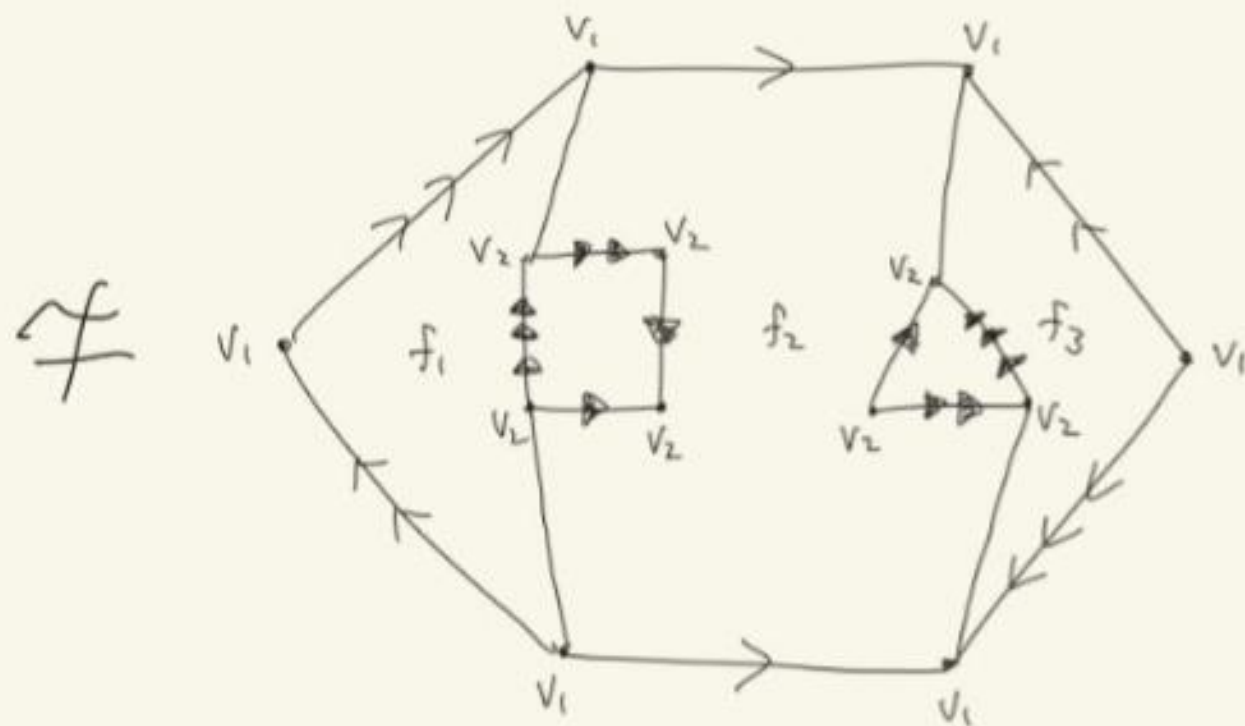




$$V - E + F = 2 - 10 + 3 = -5$$



0



-5

엄청난 사실

홀수 차원 도형은
Euler characteristic = 0

과연 좋은건가?

홀수 차원에서는 Euler characteristic으로 위상동형인지 아닌지 판별이 안됨...

2와 4가 다른 자연수임을 보이고 싶을 때

ㄱ. 짝수인지 확인해본다.

ㄴ. 4로 나누어 떨어지는지 확인해본다.

더 강력한 위상적 불변량들이 필요함

- Homology, Cohomology, Homotopy 등...

요약

1. 2차원에서는 $V - E + F = 2 - 2g$
2. $V - E + F$ 를 일반화하면 $n_1 - n_2 + n_3 - n_4 + \dots$ 이고 이를 Euler Characteristic 이라고 부른다.
3. 홀수 차원에서는 Euler Characteristic = 0, 따라서 더 강력한 위상적 불변량이 필요함

감사합니다